

3. Bayes Classification

The PDF asks for **Bayes classification using training and test data.**

Software : WEKA

Steps:

1. Open **WEKA** → **Explorer**.
2. Click **Preprocess**.
3. Open your dataset.
4. Click **Classify**.
5. Click **Choose**.
6. Select:
7. Bayes → NaiveBayes
7. Select the required **training/test option**.
8. Select the test dataset if a separate test file is provided.
9. Click **Start**.
10. Record:
 - Correctly classified instances
 - Incorrectly classified instances
 - Accuracy
 - Confusion matrix

Programming code : test.arff && train.arff

```
@relation weather
@attribute outlook {sunny,overcast,rainy}
@attribute temperature {hot,mild,cool}
@attribute humidity {high,normal}
@attribute windy {TRUE,FALSE}
@attribute play {yes,no}

@data
sunny,hot,high,FALSE,no
sunny,hot,high,TRUE,no
overcast,hot,high,FALSE,yes
rainy,mild,high,FALSE,yes
rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
rainy,mild,normal,FALSE,yes
sunny,mild,normal,TRUE,yes
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,no

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rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
rainy,mild,normal,FALSE,yes
```

OUTPUT :

The screenshot shows the WEKA Explorer interface with the NaiveBayes classifier selected. The 'Test options' section shows 'Supplied test set' selected. The 'Classifier output' section displays the following results:

```
Classifier output
=====
Error      2.0  2.0
False      4.0  3.0
[Total]    6.0  6.0

Time taken to build model: 0 seconds
Time taken to test on test set: 0 seconds
Time taken to test model on supplied test set: 0 seconds

==== Summary ====
Correctly Classified Instances      4      NO      %
Incorrectly Classified Instances    2      YES      %
Kappa Statistic                     0.7500
Mean Absolute Error                  0.2500
Root Mean Squared Error              0.5000
Relative Absolute Error              37.5000 %
Root Relative Squared Error          89.3277 %
Total Number of Instances           10

==== Detailed Accuracy By Class ====

```

	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC	ROC Area	PRC Area	Class
0.000	0.250	0.000	1.000	0.200	0.333	0.000	0.250	0.250	yes
0.750	0.000	1.000	0.750	0.800	0.778	0.000	0.875	0.833	no
Weighted Avg.	0.750	0.150	0.934	0.900	0.917	0.000	0.875	0.900	

==== Confusion Matrix ====

```

# N   0=0 Classified as
# 0 1 = yes
# 1 0 = no

```

